

Software Architecture

Architecture Context

BSc



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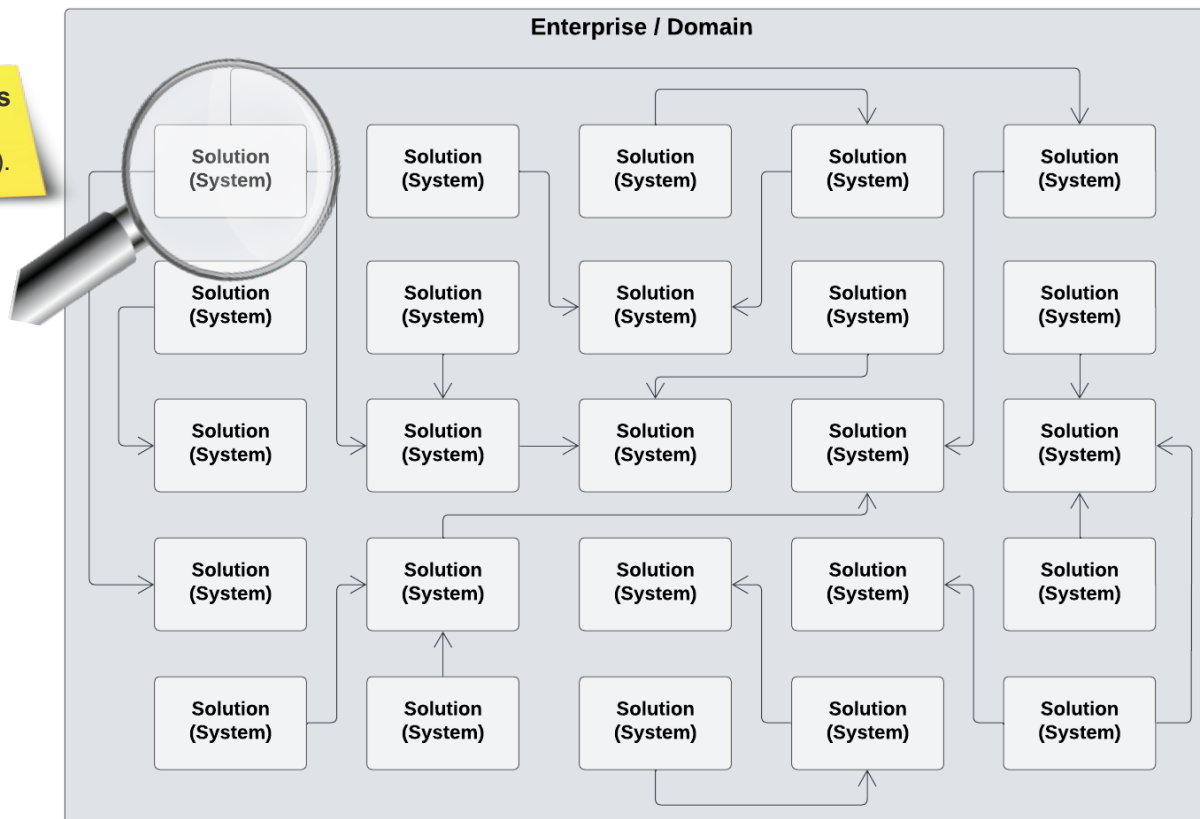
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Architecture Context

Enterprise & Domain vs. Solution & Software Architecture

Enterprise and Domain Architects view a landscape of solutions (systems) as “the system” for whose adequate architecture / design they are responsible.

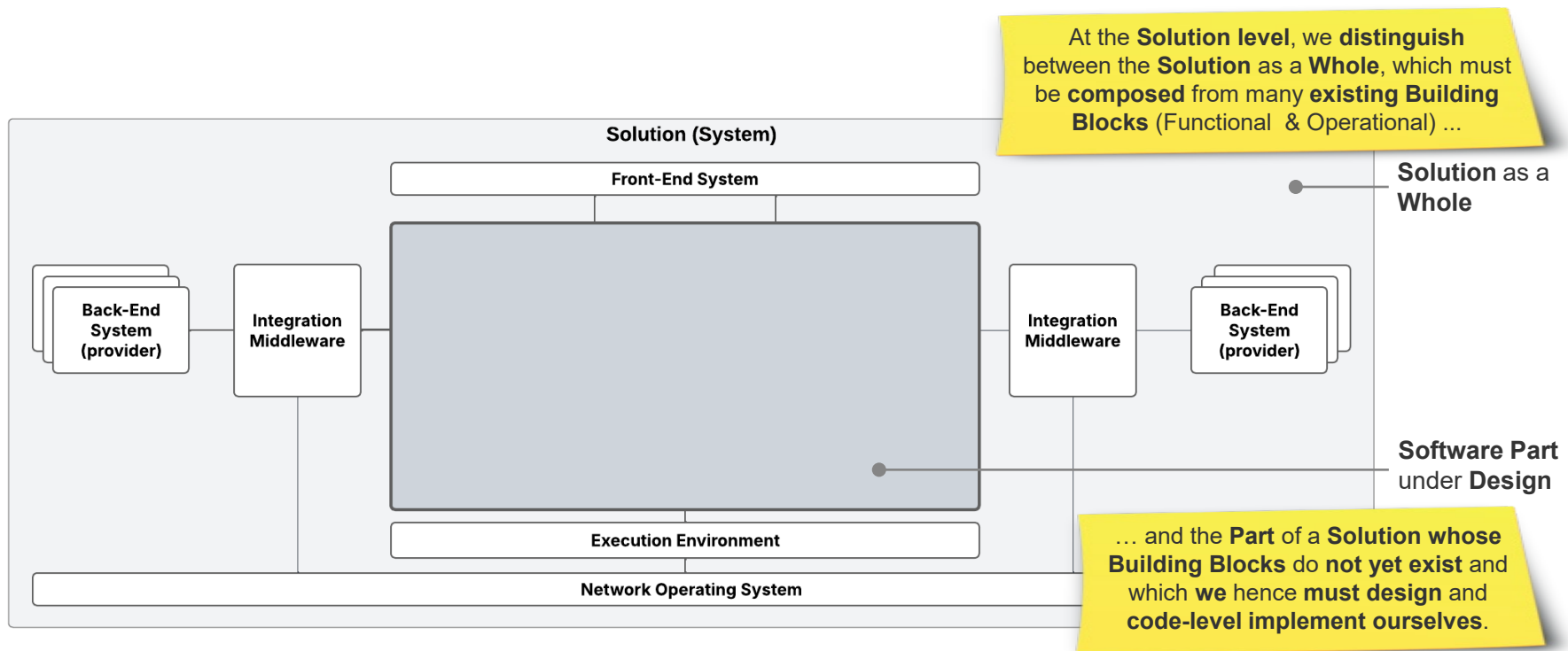
Solution & Software Architects are responsible for designing individual Solutions (Systems).



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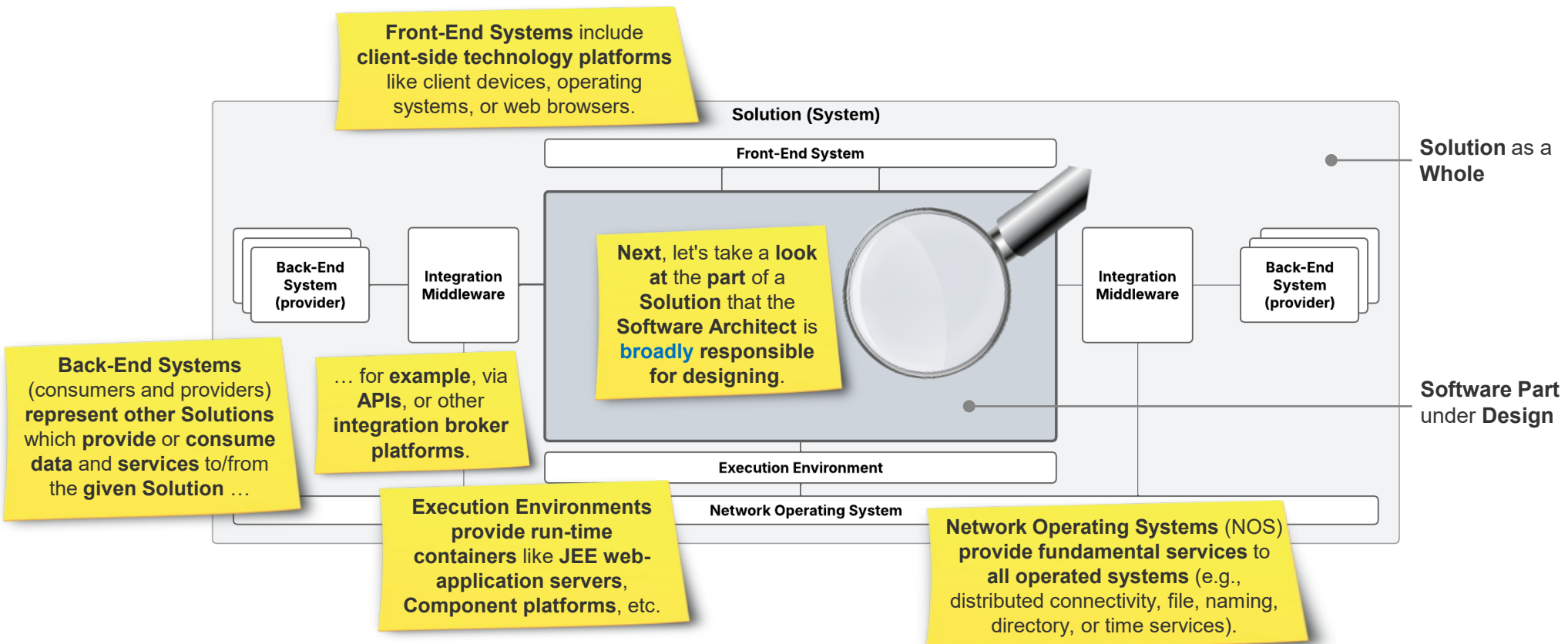
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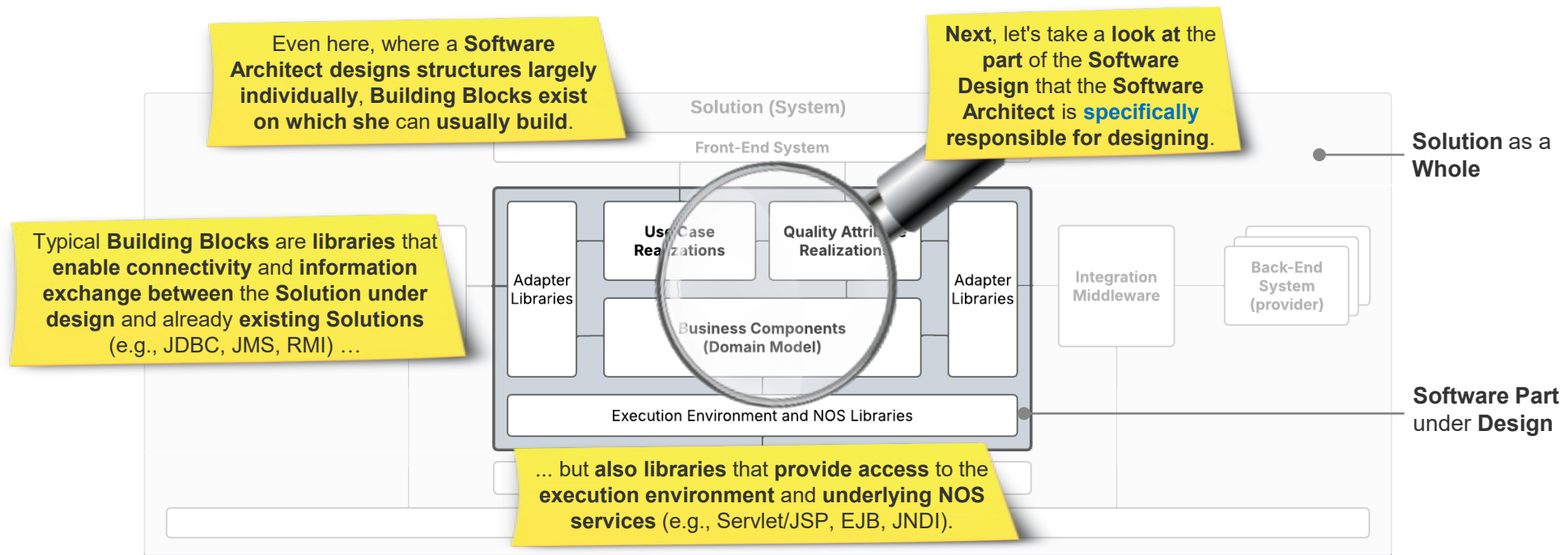
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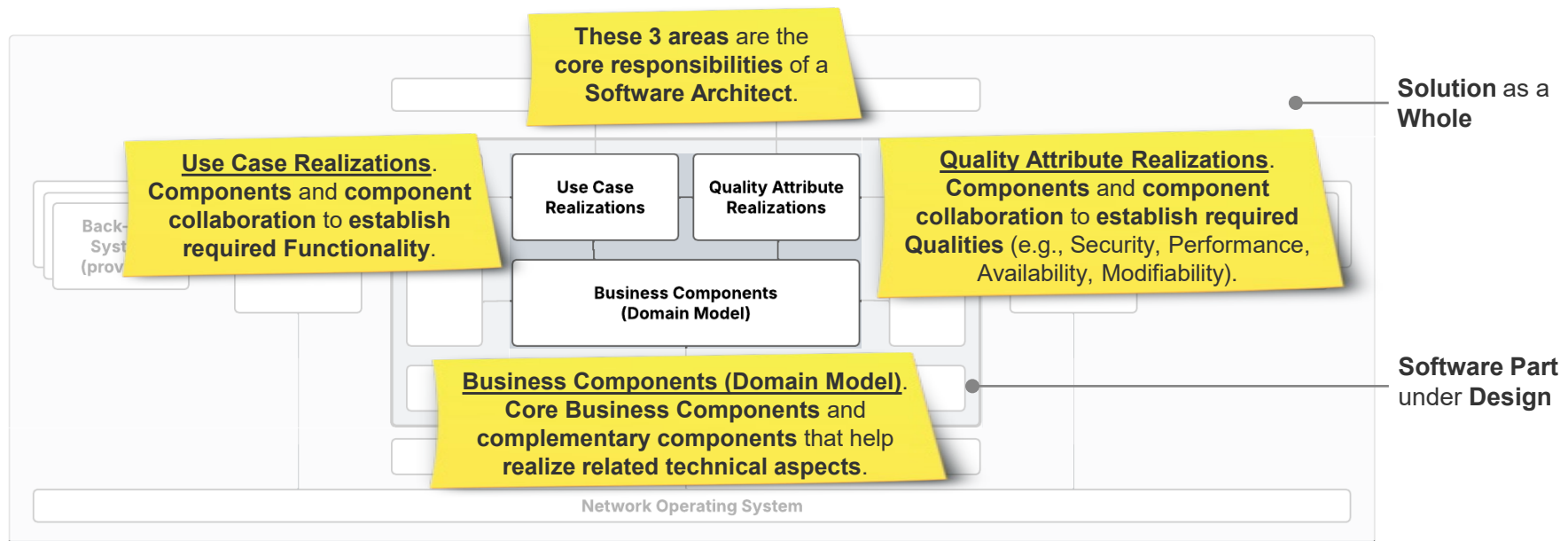
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Wrap-Up

■ Enterprise, Domain, Solution, and Software Architecture

- **Enterprise Architecture (EA) and Domain Architecture (DA) consider the entire system landscape of an organization or domain as their design object.**
 - **Enterprise Architecture: focuses on alignment between business strategy, processes, and IT capabilities.**
 - **Domain Architecture: focuses on specific business/technical domains and the architecture of the domain as a whole.**
 - **Both operate at a strategic and cross-system level.**
- **Solution Architecture (SA) focuses on the architecture of a single solution (system).**
 - **Ensures that the solution meets all functional and non-functional requirements, constraints, and assumptions.**
 - **Balances business needs, technical feasibility, and integration constraints.**
 - **Operates at a project or solution delivery level.**
- **Software Architecture (SWA) is a subset of Solution Architecture concerned with the internal structure of the software being designed and implemented.**
 - **Defines how software components are structured and interact to fulfill required functionality and quality attributes.**

Wrap-Up

- **Solution Architecture (as a whole) is composed of ...**
 - **Existing Parts**
 - **Functional or operational components already available** (e.g., existing services, databases, middleware).
 - **Examples: integration APIs, message queues, authentication systems.**
 - **Parts that need to be designed**
 - **New components developed specifically** for the **current solution**.
 - The **software architect focuses on this part** – designing the internal architecture while ensuring proper integration with existing systems.

Wrap-Up

■ Software Architect's core Responsibilities

• Use Case / Scenario Realizations

- **Designing components** and their **collaborations** to **fulfill functional requirements**.
- **Focus** on **system behavior** and **control flow** for **actor use cases** or **scenarios**.

• Quality Attribute Scenario (QAR) Realizations

- **Ensuring non-functional requirements** (e.g., performance, scalability, security, modifiability).
- **Includes architectural mechanisms** such as **caching**, **redundancy**, **monitoring**, or **abstraction**.

• Business Components (Domain Model)

- **Defining** and **structuring** the **core domain logic** and **related business components**.
- **Represents key entities**, **services**, and **rules** that **embody business knowledge**.

• **All three fields above** must be **balanced** and **interconnected** to **create a coherent software architecture**.

Questions

